***Abstract*—Robotic acquisition of tongue images for traditional Chinese medicine (TCM) tele-diagnosis requires coupling motion, on-device quality assessment, and upload policy under variable network delay. We present a cloud–edge–device stack that keeps learning-based perception out of the 1 kHz ROS control loop: a lightweight *Edge-IQA* scores each frame on the gateway, and a *LangGraph* router decides among cloud upload, edge resampling, or fail-safe abort within retry budget K. On ShezhenV3-COCO (553 test images, 500 frames $\times$ 3 seeds, baselines B0–B4), B2 reduces median RTT (M1 p50) from 374.0 ms to 208.3 ms versus naive upload (B0) while raising valid-frame rate (M2) from 0.560 to 0.604 under a documented latency model; learned B3/B4 reach M2 ≈ 0.93 with higher edge RTT. Edge-IQA on tongue ROI is calibrated on 572 validation pairs (τ_{B2}=0.465; CPU p95 $\approx$9.6 ms). Extended tables, NR-IQA baselines, stratified analyses, M6 Franka trends, and reproduction instructions are provided in the Supplementary Material submitted with this manuscript. An auxiliary Franka closed-loop track (50 trials) confirms routing decisions are reproducible on the execution stack.**

***Index Terms*—Robotics and automation, edge computing, image quality assessment, cloud–edge collaboration, medical robotics.**

Confidence-Aware Edge Routing for Robotic TCM Tongue Imaging: Edge-IQA and LangGraph Closed-Loop Acquisition

Honglei Shi and Juanjuan Zhao

I. Introduction

Tele-diagnosis for traditional Chinese medicine (TCM) tongue inspection needs repeatable, high-quality images before cloud analytics and electronic medical records integration [1], [2]. Clinical diagnosis relies on color, texture, coating, and shape cues that degrade under motion blur, uneven lighting, and incorrect working distance. Naive pipelines upload every frame, wasting bandwidth on blurred or poorly exposed captures and inflating round-trip time (RTT) on slow links. *Garbage-in garbage-out* effects are acute: a single blurred upload triggers wasted RTT and may propagate low-confidence features into downstream classifiers. Edge-side gating must be (i) fast for closed-loop capture, (ii) interpretable for audit, and (iii) calibratable on real tongue corpora rather than synthetic placeholders alone.

We study a **cloud–edge–device** stack for a Franka Research 3 (FR3) platform with ROS 2 and MoveIt [3], [4]. Perception and routing live in a Python edge gateway (`franka_api_server`) and companion edge modules (`edge_iqa`, `langgraph_router`)—not inside hardware plugins—so algorithms remain testable without a robot. A lightweight *Edge-IQA* fuses Laplacian sharpness and exposure into $Q_{\mathrm{img}} \in [0, 1]$ on COCO tongue ROI crops; a LangGraph router decides `upload_cloud`, `resample_edge`, or `fail_safe` under retry budget $K{=}2$. An optional learned baseline B3 fine-tunes MobileNetV3-Small on the training split for comparison.

We validate on public **ShezhenV3-COCO** (6,719 images with bounding boxes): τ is calibrated on 572 validation pairs (clear + synthetic blur) with ROI enabled, and the main matrix replays 500 test frames per seed with real Edge-IQA and B3 scores from 553 held-out images.

Contributions: (1) Edge-IQA with tongue ROI—$O(HW)$ CPU scoring, interpretable blur/exposure flags, and p95 below 10 ms; (2) auditable LangGraph three-branch routing logged as JSONL for clinical audit trails; (3) dual-track evaluation—offline baselines B0–B4 on ShezhenV3 (3 seeds) plus an auxiliary Franka closed-loop track (M6) confirming execution-stack feasibility without altering Table VI.

Section II reviews related work; §III–IV present the method and ShezhenV3 evaluation; §V concludes.

Manuscript received Month DD, 2026; revised Month DD, 2026.

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II. Related Work

a) Robotic medical imaging.: ROS 2 and MoveIt enable repeatable manipulator-mounted cameras [3], [4]. We adopt named *skills* (`go_to_tongue_pose`, `go_to_face_pose`) instead of online impedance tuning, matching clinical repeatability while keeping Paper I independent of force-control threads.

b) Image quality assessment.: **Full-reference** metrics (PSNR, SSIM) require pristine references unavailable per patient. **Hand-crafted NR-IQA** (Laplacian, BRISQUE, NIQE [5]) runs on CPU but seldom exposes blur/exposure semantics for audit logs. **Deep NR-IQA** (HyperIQA [6], CLIP-IQA [7]) improves MOS correlation at GPU cost and limited interpretability. Edge closed-loop capture favors linear-time operators with auditable failure modes; Edge-IQA follows this line with explicit flags and calibrated τ (Table I).

c) Edge–cloud orchestration.: Split inference reduces bandwidth in mobile vision [8], [9]; Zhou et al. [9] taxonomize edge intelligence tiers. We gate *upload decisions* before heavy cloud models execute, using a lightweight scorer (p95 ≈9.6 ms CPU) rather than partial DNN migration. LangGraph-style routers express branching policies with logs; ReAct [10] motivates interleaving tool actions with routing.

d) TCM tongue analysis and positioning.: Cloud TCM pipelines [1], [2] assume usable inputs; ShezhenV3-COCO supplies boxes for *downstream* classifiers. Paper I addresses *acquisition quality and routing* upstream on the full public corpus. Compared with generic edge offloading, we target robotic *capture* loops where quality varies shot-to-shot; compared with NR-IQA literature, we prioritize thresholdable Q_{img} and JSONL auditability aligned with medical device software trends.

III. Method

A. System Architecture

Fig. 1 shows the stack. **Device layer:** FR3 manipulator, RGB camera, and ROS 2/MoveIt PTP execution; named skills invoke calibrated joint poses via the edge gateway (`PAPER1_MODE=1` disables stiffness routes in experiments). **Edge layer:** FastAPI gateway (`franka_api_server`) hosts Edge-IQA and LangGraph; `sim/franka_bridge` connects graph nodes to `/vision/evaluate` and skill endpoints. **Cloud layer:** higher-level vision and EMR integration (stub); upload occurs only when edge quality and routing accept the frame.

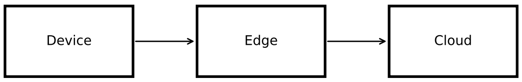

Fig. 1. Cloud–edge–device stack for robotic tongue-image acquisition.

TABLE I
IQA APPROACHES FOR EDGE ACQUISITION GATING (QUALITATIVE).

Method	Lat.	GPU	Flags	τ	Audit
Edge-IQA	$\sim$10 ms	No	Yes	Yes	Yes
Laplacian	$\sim$5 ms	No	Part.	Yes	Yes
NIQE/BRISQUE	50–200 ms	No	Lim.	Map	Wrap.
Deep NR-IQA	10–100+ ms	Often	No	Map	Svc.
B3 (learned)	15–40 ms	Opt.	No	Yes	Sep.

Algorithm 1 Edge-IQA scoring

Resize I to 512×512, convert to grayscale
$S \leftarrow \mathrm{normalize}(\mathrm{Var}(\nabla^2 I))$
$E \leftarrow 1 - |\mathrm{mean}(I) - 0.5|/0.5$
$Q_{\mathrm{img}} \leftarrow 0.65S + 0.35E$; derive flags from (S, E)
return Q_{img}, flags

Each trial logs ISO 8601 timestamps (`t_capture`, `t_iqa`, `t_route`) and decisions to JSONL; main RTT metrics are computed offline by `run_matrix_tcm.py` on ShezhenV3 test images. Offline RTT aggregates a documented latency model: capture 5 ms, IQA $\mathcal{N}(9, 2^2)$ ms, routing 1.5 ms, resample p50≈50 ms, cloud upload 50–550 ms (see `sim/NETEM.md`). Relative baseline ordering is the primary claim; ShezhenV3 blur labels are synthetically injected Gaussian defocus when human quality annotations are unavailable.

B. Edge Image Quality Assessment

Edge-IQA scores each frame on CPU with $O(HW)$ cost on a 512×512 resize. Let $S = \mathrm{Var}(\nabla^2 I)$ normalized to $[0, 1]$ and E an exposure term peaked at mid-gray:

$$Q_{\mathrm{img}} = 0.65\,S + 0.35\,E. \tag{1}$$

Binary flags mark `blur` ($S < \tau_s$), `underexposed`, or `overexposed` means. Robotic tongue capture fails mainly from motion blur and exposure drift; a single threshold τ gates LangGraph routing. On ShezhenV3 validation with COCO tongue ROI, clear/blur medians are 0.516 and 0.414 (τ_{B2}=0.465); CPU p95 is 9.6 ms (`edge_iqa/scorer.py`). The gateway spawns `python -m edge_iqa.cli --json` in a subprocess to avoid GIL contention with `rclpy`. Optional baseline B3 fine-tunes MobileNetV3-Small [11] on the train split; the primary deployable path is interpretable B2.

TABLE II
EDGE-IQA CALIBRATION: FULL FRAME VS. TONGUE ROI (VAL, n=572 PER COHORT).

Setting	Med. clear	Med. blur	τ	Sep.
Full frame	0.564	0.431	0.498	−0.510
Tongue ROI	0.516	0.414	0.465	−0.541

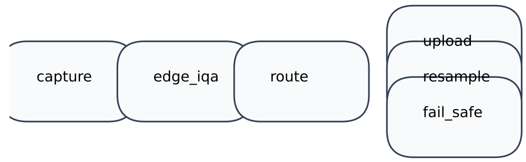

Fig. 2. LangGraph confidence-aware routing flow.

C. COCO Tongue ROI Cropping

ShezhenV3-COCO provides COCO boxes $[x, y, w, h]$; we union multi-instance annotations, pad 8%, and crop before resize (`edge_iqa/coco_roi.py`). Full-frame scoring includes cheeks and background Laplacian energy unrelated to tongue focus; ROI aligns the quality signal with downstream TCM classifiers.

Table II compares calibration settings. Minimum-clear minus maximum-blur separation remains negative in both settings, so τ_{B2}=0.465 is a cohort operating point—we report routing outcomes and Spearman ρ (M5, ρ≈0.36–0.47) rather than perfect separability. When COCO boxes are unavailable at runtime, Edge-IQA falls back to full-frame scoring (`use_roi=false`).

D. Confidence-Aware Routing

Given score Q_{img} and retry counter r, the router implements:

$$\mathrm{route}(s) = \begin{cases} \text{upload\_cloud}, & Q_{\mathrm{img}} \geq \tau \land r \leq K, \\ \text{resample\_edge}, & Q_{\mathrm{img}} < \tau \land r \leq K, \\ \text{fail\_safe}, & r > K. \end{cases} \tag{2}$$

with K=2 in our experiments. LangGraph nodes `capture` → `edge_iqa` → `route` branch to resample, upload, or abort; `TrialState` carries `q_img`, `flags`, `retry_count`, and ISO-8601 timestamps. Routing overhead (≈3.6 µs/trial on 1,000 trials) is negligible versus the 9 ms IQA step; LangGraph additionally provides checkpoint persistence and graph visualization (Fig. 2).

Each `resample_edge` applies *physical skill transforms* before re-scoring: `adaptive_visual_damping` reduces Gaussian blur radius on the clear reference; `exposure_loop_tuning` corrects gain/bias when exposure flags fire—no stochastic score inflation. JSONL logs ten fields per frame for audit (`trial_id`, `q_img`, `flags`, `route_decision`, timestamps, `latency_ms`).

IV. EXPERIMENTS

A. Experimental Setup

We evaluate offline baselines B0–B4 on the ShezhenV3-COCO test split [1] (553 images; validation n=572 for τ;

Algorithm 2 Confidence-aware routing loop (B2)

$r \leftarrow 0$; capture frame I
repeat
  $(Q, \texttt{flags}) \leftarrow \text{EdgeIQA}(I)$
  $d \leftarrow \text{Route}(Q, r, \tau, K)$ {Eq. 2}
  if $d = \texttt{upload\_cloud}$ **then**
    break
  else if $d = \texttt{resample\_edge}$ **then**
    $r \leftarrow r + 1$; recapture I
  else
    fail_safe; **break**
  end if
until $r > K$

TABLE III
SHEZHENV3-COCO ANNOTATION COUNTS BY TONGUE CONDITION (ALL SPLITS).

Code name	Gloss	#Ann.
`baitaishe`	White-coating tongue	5040
`hongdianshe`	Red-spot tongue	3653
`liewenshe`	Fissured tongue	1886
`chihenshe`	Tooth-marked tongue	1482
`hongshe`	Red tongue	1466
`huangtaishe`	huangtaishe	1119
`pangdashe`	Chubby tongue	689
`botaishe`	Peeling tongue	581
`shenquao`	shenquao	495
`xinfeiao`	xinfeiao	372
`gandanao`	gandanao	292
`shoushe`	Thin tongue	285
`huataishe`	huataishe	283
`zishe`	Purple tongue	231
`piweiao`	piweiao	186
`heitaishe`	heitaishe	122
`jiankangshe`	Healthy tongue	21
`gandantu`	gandantu	9
`shenqutu`	shenqutu	2
`xinfeitu`	xinfeitu	2

train n=5,594 for B3; total 6,719 with COCO boxes). Table III summarizes diagnostic morphology labels; conditions describe tongue pathology rather than capture quality, motivating our separate Edge-IQA blur/exposure axis. **B0**: cloud upload only; **B1**: Edge-IQA gate without resampling; **B2**: Edge-IQA + LangGraph with physical resampling (proposed); **B3/B3t/B4**: learned MobileNet and hybrid Tier-A/B gates (Table VII).

Each test frame uses real `scorer.py` scores; with probability 0.5 we use the original capture (clear cohort) and otherwise apply Gaussian blur ($r \in [2.5, 5.0]$ px)—mirroring validation calibration. We simulate 500 frames per seed ($\{0, 1, 2\}$) under the latency model in §III-A; metrics are M1 (RTT p50/p95), M2 (valid-frame rate), and M3 (retry rate). All main-text numbers are produced by `run_matrix_tcm.py` reading `recommended_tau.json`; auxiliary Franka closed-loop trials (M6, 50 trials) confirm routing trends but do not alter Table VI.

TABLE IV
SHEZHENV3-COCO TONGUE DATASET SPLITS USED IN PAPER I.

Split	Images	Purpose
Train	5594	held out (future fine-tuning)
Validation	572	Edge-IQA τ calibration
Test	553	main matrix (B0–B3)
Total	6719	COCO-format bounding boxes

TABLE V
EDGE-IQA CALIBRATION ON SHEZHENV3 VALIDATION SPLIT ($n = 572$ CLEAR + $n = 572$ SYNTHETIC BLUR).

Metric	Clear cohort	Blur cohort
Median Q_{img}	0.516	0.414
Mean Q_{img}	0.519	0.414
Std	0.116	0.102
Routing threshold τ	0.465	
Spearman ρ (M5)	0.335 ($n = 1144$)	

Key hyperparameters: τ_{B2}=0.465 (ROI), K=2, sharpness/exposure weights $0.65/0.35$, resize 512^2, blur $r \in [2.5, 5.0]$ px, τ_{B3}=0.5 for learned gates.

B. Edge-IQA Validation

On the validation split (n=572 clear + synthetic blur pairs), Fig. 3 shows overlapping Q_{img} histograms. Table V lists cohort medians (0.516/0.414), deployed τ_{B2}=0.465, and negative separability (-0.541): some blurred validation crops score above the weakest clear crops, so τ is a cohort-median operating point rather than a hard separator. Spearman $\rho\approx$0.36–0.47 on real texture motivates pairing interpretable B2 with learned comparators (Table VII).

Each test frame is scored by the same `scorer.py` invoked offline; latency components follow §IV-A. This ensures Table VI reflects *real* quality scores while preserving the reproducible closed-loop simulator for RTT aggregation. An auxiliary Franka track (M6, 50 trials on fake hardware) logs JSONL through the live gateway stack and confirms routing decisions are reproducible on the execution path; M6 RTT trends are reported in the online supplement and must not be merged into Table VI.

C. Main Results

Table VI summarizes B0–B2 (mean $\pm$ std over three seeds; frame-level bootstrap 95% CIs in caption). B2 improves M2 from 0.560 to 0.604 and lowers M1 p50 to 208.3 ms versus B0; B1 keeps M2 at 0.560 because sub-threshold frames still reach the cloud without closed-loop resampling. B2 raises M3 to 0.440 because physical resampling consumes retry budget before upload.

Learned B3/B4 reach M2 $\approx$ 0.93–0.97 with M1 p50 $\approx$340–350 ms (Table VII); we report them as comparators, not the primary deployable path. Table VIII decomposes M2 by clear-source vs. blur-injected cohorts: B2 gains concentrate on blur-injected frames (0.373$\rightarrow$0.460) where resampling recovers quality before upload.

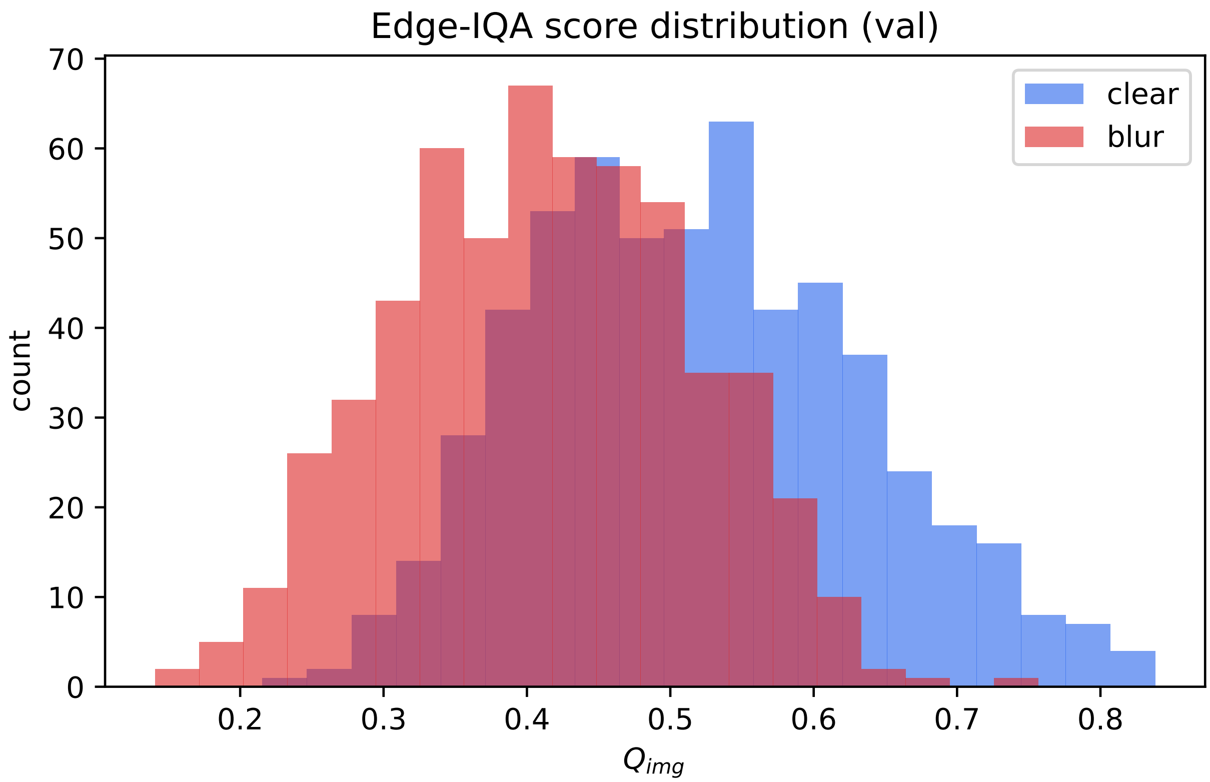

Fig. 3. Edge-IQA Q_{img} on ShezhenV3 validation: clear vs. synthetically blurred pairs (n=1,144). Overlap motivates cohort-median τ rather than hard separation.

TABLE VI
MAIN RESULTS (SHEZHENV3 TEST, 553 IMAGES; EDGE-IQA, PHYSICAL RESAMPLE, τ_{B2}, 3 SEEDS). CELLS REPORT MEAN ± STD OVER SEEDS. FRAME-LEVEL BOOTSTRAP 95% CIS FOR M2 (POOLED OVER 3 × 500 FRAMES): B0 M2=0.560 [0.535,0.585]; B1 M2=0.560 [0.535,0.585]; B2 M2=0.604 [0.579,0.628].

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B0	374.0 ± 7.9	656.9 ± 2.9	0.560 ± 0.007	0.000 ± 0.000
B1	322.7 ± 15.6	534.1 ± 1.2	0.560 ± 0.007	0.000 ± 0.000
B2	208.3 ± 2.8	526.9 ± 2.3	0.604 ± 0.005	0.440 ± 0.007

TABLE VII
LEARNED IQA AND HYBRID GATING (SHEZHENV3 TEST, PHYSICAL RESAMPLE, 3 SEEDS).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry rate
B3 (MobileNet, τ_{B2}=0.465)	345.6	606.8	0.934	0.481
B3 (MobileNet, τ_{B3}=0.5)	341.4	609.6	0.929	0.481
B4 (Hybrid Tier-A/B)	343.1	617.7	0.936	0.440

Fig. 4 summarizes the M2–M1 trade-off across B0–B4; Fig. 5 shows RTT CDFs where B2 shifts mass left relative to B0. Fig. 6 reports M2 across seeds; low variance (std ≤0.007) reflects deterministic test-set scores given a shared frame plan per seed.

a) Takeaway.: On a full public TCM test cohort, B2 with physical skill transforms and auditable LangGraph logs remains the primary deployable path; Hybrid B4 combines interpretable exposure gating with learned blur ranking when higher M2 justifies extra edge RTT.

TABLE VIII
STRATIFIED M2 (VALID-FRAME RATE) BY INJECTION COHORT; POOLED OVER 3 SEEDS × 500 FRAMES.

Baseline	M2 (clear-source)	M2 (blur-injected)
B0	0.733 (779 frames)	0.373 (721 frames)
B2	0.737 (779 frames)	0.460 (721 frames)

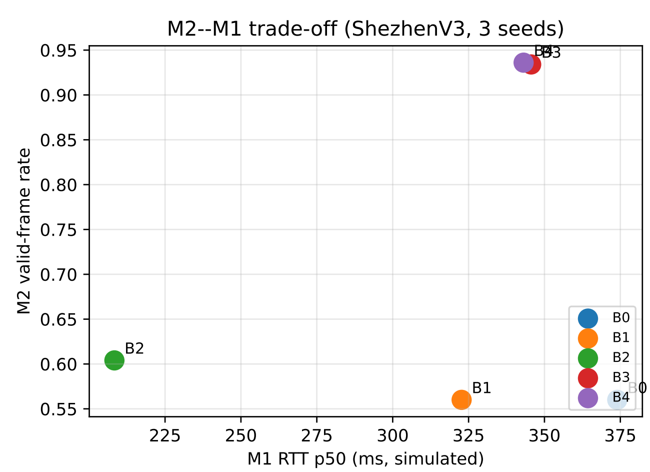

Fig. 4. M2 vs. simulated M1 p50 (mean over 3 seeds). B2 favors low RTT; B3/B4 favor high M2 at higher edge cost.

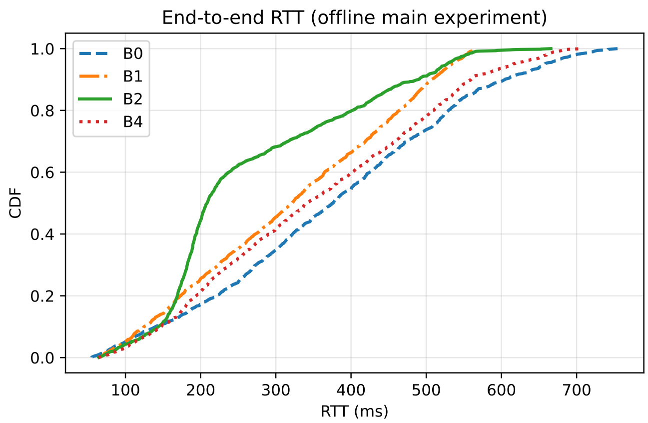

Fig. 5. End-to-end RTT CDF under the simulated latency model (3 seeds).

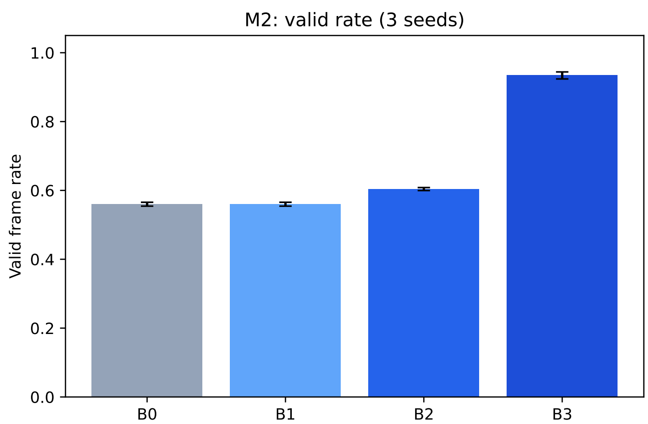

Fig. 6. Valid frame rate (M2) per baseline; error bars = std over seeds.

TABLE IX
QUICK ABLATION OF ROUTING EXTENSIONS (SEED 0, n=60, PHYSICAL RESAMPLE).

Baseline	M1 p50	M2	M3
B2 (fixed τ)	211.3	0.600	0.417
B2d (dynamic τ_t)	209.0	0.550	0.467
B2m (EMA fusion)	280.4	**0.650**	0.417
B2u (utility U)	205.4	**0.650**	0.367

D. Ablation

Figure 7 sweeps routing threshold τ for B2 (500 frames, seed 0). The deployed main matrix uses τ_{B2}=0.465 from ROI validation; τ_{full}=0.498 appears in the full-frame ROI ablation (Table II).

Table IX compares fixed-threshold B2 against dynamic τ_t (B2d), multi-frame EMA (B2m), and utility routing (B2u) on 60 test frames (seed 0). B2m and B2u raise M2 from 0.60 to 0.65 at the cost of extra edge compute or stricter upload utility; full 3-seed sweeps are in the online supplement.

E. Discussion

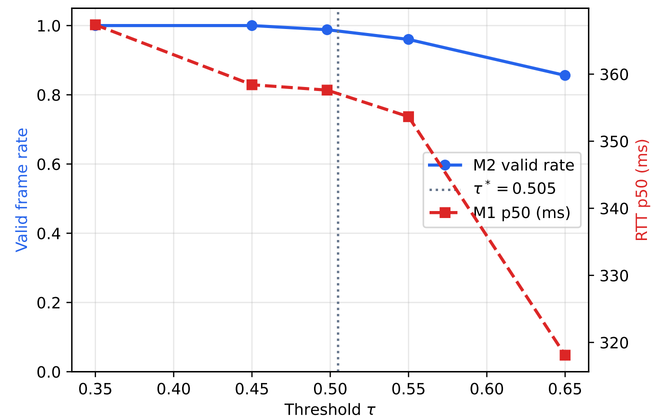

Fig. 7. Ablation over τ: valid rate and RTT p50 for B2 on TCM test frames.

Table VI reports *relative* baseline comparison under the documented simulated latency model (§IV-A); absolute RTT values are not live hospital traces. B2 reduces M1 p50 from 374.0 ms (B0) to 208.3 ms and raises M2 from 0.560 to 0.604 on ShezhenV3 test frames with physical resampling and ROI Edge-IQA (τ_{B2}=0.465). All three seeds show B2 outperforming B0 on M1 and M2; low M2 variance reflects deterministic test-set scores given a shared frame plan, with remaining randomness from the latency model and frame-plan shuffling.

COCO tongue ROI (8% padded union box) aligns scoring with downstream anatomy but does not eliminate clear/blur overlap on Laplacian features (Table II, negative separation -0.541). Because ShezhenV3 lacks human blur labels, we inject Gaussian defocus on half of test frames, mirroring validation calibration; M5 Spearman $\rho\approx$0.36–0.47 reflects residual overlap under real texture. B3 (MobileNetV3-Small, 98.3% val accuracy) achieves M2= 0.934 at τ_{B2} versus B2 M2= 0.604; we retain B2 for lowest M1 p50 and auditable `flags`, while B3/B4 suit higher valid-rate targets with extra edge compute.

Edge-IQA and LangGraph are hardware-agnostic: any edge gateway exposing HTTP `/vision/evaluate` and skill endpoints can replay the same JSONL protocol. M2 measures frames passing the scorer threshold—a routing proxy, not independent clinical quality annotation; future work should correlate M2 with downstream classifier confidence on accepted vs. rejected frames. Extended tables (NR-IQA baselines, per-seed breakdown, deployment notes, failure modes) are provided in the Supplementary Material.

V. CONCLUSION

We presented Edge-IQA and LangGraph confidence-aware routing for robotic TCM tongue-image acquisition on a Franka ROS 2 stack, validated end-to-end on ShezhenV3-COCO (6,719 images). Keeping IQA and routing outside the 1 kHz control loop preserves real-time safety while enabling JSONL-audited closed-loop policies. On the held-out test split with real ROI Edge-IQA scores, B2 improves median RTT and valid-frame rate versus naive upload (M1 p50 208.3 vs. 374.0 ms; M2 0.604 vs. 0.560) at τ_{B2}=0.465; M5 $\rho\approx$0.36–0.47 characterizes overlap under real tongue texture.

Limitations. (1) Validation and test blur are synthetically injected Gaussian defocus, which may under-represent directional motion smear. (2) Edge-IQA overlap between clear and blurred cohorts motivates learned refinement (B3/B4) alongside interpretable B2. (3) M2 is scorer-internal; end-to-end clinical validation correlating gating with downstream classifier performance remains future work. (4) Three random seeds provide consistent trends; bootstrap CIs in Table VI support descriptive claims but larger seed budgets would strengthen formal inference. (5) Cloud vision and EMR integration remain stubs; M6 confirms execution-stack feasibility but does not replace live clinical deployment data.

Future work. On-device tongue detection for ROI without COCO boxes, Gazebo camera injection, NR-IQA baselines under identical resampling protocols, and multi-center clinical pilots re-calibrating τ_{B2} on 50+ local sessions before site-specific upload policies.

Acknowledgments

The authors used generative AI tools for English prose editing; all metrics were verified against `experiments/results/*.json`. This work was supported by Taiyuan University of Technology; no competing interests are declared. ShezhenV3-COCO is public; code, splits, and reproduction steps are provided in the Supplementary Material. No new human subjects were recruited; JSONL logs contain no patient identifiers.

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